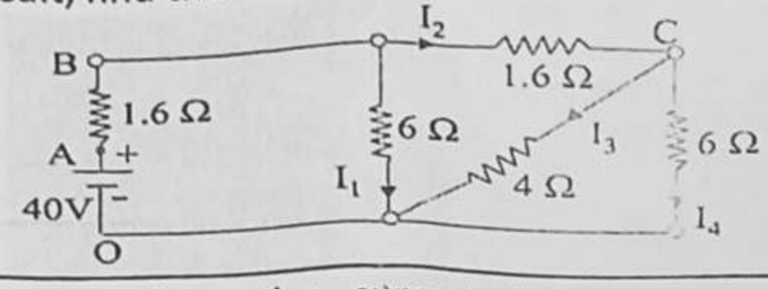
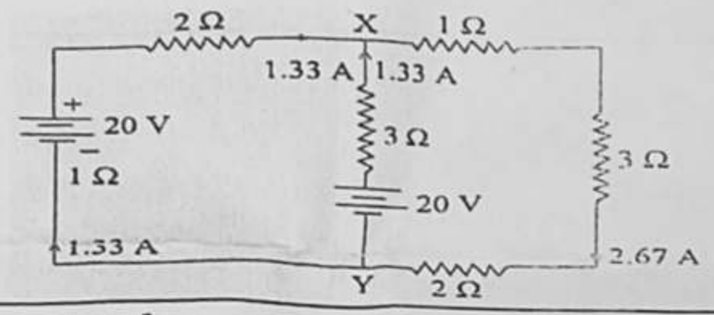
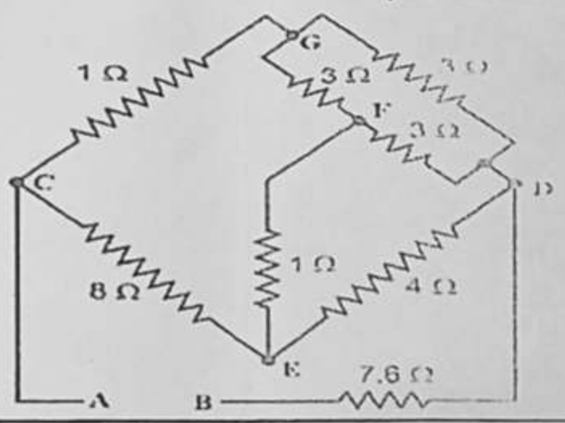
MID-TERM EXAMINATION
B.Tech (CSE/ECE/ECE-AI)
1st semester (October, 2023)
OFFLINE mode

Subject Code: BEC-110	Subject: Basic Electrical Engineering
Time: 1 ½ Hours	Maximum Marks: 30
Note: Q. 1 is compulsory. Attempt any two questions from the rest.	

Q1		(2.5*4)	
	(a) State Reciprocity theorem with example.		
	(b) Three resistors 6Ω, 15Ω and 18Ω are connected in star. Determine the resistances for an equivalent delta connection?		
	(c) For the given circuit, find the current through all resistances. 		
	(d) The equation of an alternating current i is i=42.42 sin628t. Find (i)Maximum value of current (ii)frequency & (iii) form factor.		
Q2	(Attempt any two Parts) UNIT-1	(5,5)	
	(a) State & Prove Superposition theorem. Also, calculate the current flowing in the branch XY. 		
	(b) Explain Star to Delta transformations. Find equivalent resistance across A and B: 		
	(c) State and Prove Maximum Power Transfer Theorem.		
Q3	(Attempt any two Parts) UNIT-2	(5,5)	
	(a) A series circuit consists of a resistance of 6Ω, an inductance of 500mH and a variable capacitance connected across a 100V, 50Hz supply. Calculate (i) The capacitance required to produce a series resonance condition (ii) The voltages generated across both the inductor and the capacitor at the point of resonance. (iii) Bandwidth (iv) Quality Factor (v) Power Consumed in the circuit.		
	(b) A pure resistance of 50 Ω is connected is in series with a pure capacitor of 100 μF. The series combination is connected across 100 V, 50 Hz supply. Find (a) Impedance (b) Current (c) Power Factor (d) Voltage across R and C.		
	(c) Describe Bandwidth, Impedance, Q-factor and resonance in R-L-C series circuit.		

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